Soumik Sahoo

B.Tech in Engineering Physics

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Research Interests

My research interests center on **semiconductor spin qubits** and **superconducting qubits**, particularly their **design, control, readout, and coherence**. I am interested in experimental quantum-device physics and in understanding how device materials and architecture shape qubit performance and scalability. I am particularly interested in how **experimental characterization** can inform improvements in **qubit fidelity and reliable operation**.

Degree/Examination	Board/Univ.	Institute	Year	CPI/%
B.Tech in **Engineering Physics**	IIT Bombay	IIT Bombay	2023–2027	8.9/10
Higher Secondary (**XII**)	CHSE, Odisha	Saraswati Vidya Mandir, Neelakanthanagar	2022	89.66%
Secondary (**X**)	BSE, Odisha	Saraswati Vidya Mandir, Singhpur	2020	93.50%

- Pursuing an **Honors** in **Physics** (Department of Physics, IIT Bombay)

Scholastic Achievements

- Achieved **99.87 percentile** among 1.2 million candidates in JEE Main *(2023)*
- Secured **98.2 percentile** among 0.2 million candidates in JEE Advanced *(2023)*

Research Experience

- **Charge Sensing and Qubit Characterization of Gate-Defined Quantum Dots** BTP-1 *(Aug '25 – Present)*
  B.Tech Thesis: Prof. Suddhasatta Mahapatra & Prof. Uditendu Mukhopadhyay | Department of Physics, IIT Bombay
  - Characterized gate-defined quantum-dot devices at millikelvin temperatures, starting from initial measurements to forming SET and observing Coulomb oscillations
  - Tuned neighbouring dots into a double quantum dot using single-electron-transistor charge sensing, mapping its charge-stability diagram and tracking charge transitions with high sensitivity
  - Extracted interdot tunnel couplings by fitting charge-sensor data across the interdot transition to a two-level model, a key parameter for exchange-based spin-qubit operation
  - Developing pulse sequences to measure the spin relaxation time (T_1), and contributing to the lab's RF-reflectometry setup for fast, single-shot readout
  - Operate a dilution refrigerator (sample loading to cooldowns) and measure with QCoDeS
- **Induced Superconductivity in Magnetic Topological-Insulator Thin Films** Draft *(Summer '26 – Present)*
  Research Internship: Prof. Thomas Schmidt | University of Luxembourg | Manuscript in preparation
  - Developed a Green's-function theory of superconductivity induced in magnetic topological-insulator (MTI) thin films by an s-wave superconductor, deriving normal and anomalous propagators to fourth order in tunneling and proving that odd orders vanish by fermion parity
  - Evaluated kernels by k_z contour integration and resummed all even orders via the Dyson equation into a frequency-dependent 8×8 BdG Hamiltonian, revealing rank-one, orbital-selective pairing
  - Mapped the BdG Chern-number phase diagram, fixing the Chern number from Γ-point states via a rotation symmetry; cross-checked with Z2Pack Wilson loops and the Fukui–Hatsugai–Suzuki method
  - Identified the chiral topological-superconductor phase ($|N| = 1$, chiral Majorana edge modes).
- **Interaction-Induced Caging in Non-Hermitian Lattices** Report *(Jan '26 – Apr' 26)*
  Supervisor: Prof. Lee Ching Hua | Department of Physics, National University of Singapore
  - Derived the generalized Brillouin zone of the Hatano–Nelson chain and its skin-mode localization length, $\kappa = \ln(t_L/t_R)/2$, confirmed numerically by root-tracking of the characteristic polynomial
  - Built a 2D hybrid SSH–Hatano–Nelson Hamiltonian hosting a spectrally isolated corner trap, and computed the spectral winding number ($W = 1$) that protects the skin modes

- Built the Bose–Hubbard Hamiltonian in Fock space, reproduced the non-monotonic interaction-driven caging and de-caging of bosonic clusters, and mapped the hard-core limit to a non-Hermitian XXZ chain

- **Free Fermions in Disguise** Report *(Aug '25 – Dec '25)*
  Supervised Learning Project: Prof. Sumiran Pujari | Department of Physics, IIT Bombay
  - Analyzed Fendley's spin chain built purely from four-fermion terms, which no Jordan–Wigner transformation reduces to bilinear form yet which has an exact free-fermion spectrum
  - Constructed its conserved charges, commuting transfer matrices and explicit free-fermion raising and lowering operators for the open chain, accounting for the exponential degeneracies of the spectrum
  - Verified the spectra numerically up to $M \approx 8$ and extended the approach to the transverse-field Ising model

- **Perturbation Theory for the Fractional Quantum Hall Effect** Report *(2024 – 2025)*
  Supervisor: Prof. Prashant Kumar | Department of Physics, IIT Bombay
  - Studied a perturbative route to the FQH effect that expands about the integer state in continuously attached flux m, motivated by the universality class of the FQH–insulator transition
  - Derived the n-particle distribution functions of the integer state via the plasma analogy (2D one-component plasma at $\Gamma = 2$) as determinantal kernels, the inputs needed at each order

- **Electronic Transport and Scattering in Conductors** *(Jan '25 – Apr '25)*
  Supervisor: Prof. Soumya Bera | Department of Physics, IIT Bombay
  - Modeled impurity and Umklapp scattering via the Boltzmann equation (relaxation-time approximation)
  - Derived temperature-dependent resistivity from electron–phonon and electron–electron interactions

- **Non-Equilibrium Green's Function Formalism** Link | *Computational Physics Project* *(Dec '24 – Jan '25)*
  - Implemented the NEGF formalism to model coherent electron transport in nanoscale devices, computing I–V characteristics and reproducing conductance quantization
  - Coupled the NEGF solver to a self-consistent-field loop with band-structure and subband modeling, computing density of states, local density of states, potential profiles, and transmission functions

Course Projects

- **Spectra of Kitaev Chains** Presentation *(PH 554)* *(Jan '26 – Apr' 26)*
  Prof. Soumya Bera | Department of Physics, IIT Bombay
  - Formulated finite Kitaev chains in both the many-body Fock basis and the quadratic BdG (Nambu) basis, comparing Hilbert-space scaling, Hamiltonian structure, and low-energy spectra
  - Computed Zeeman-dependent spectra, topological invariants, and finite-size scaling to distinguish true from false poor man's Majorana modes under uniform and site-dependent fields

- **Chiral Tunnelling and the Klein Paradox in Graphene** Presentation *(PH 436)* *(Aug '25 – Nov '25)*
  Prof. Hridish Pal | Department of Physics, IIT Bombay
  - Derived relativistic reflection and transmission amplitudes across electrostatic barriers, using wave-vector matching and group-velocity corrections to resolve the Klein paradox at oblique incidence
  - Built graphene's tight-binding and Bloch Hamiltonians near the Dirac points, linking pseudospin conservation and chirality to perfect normal-incidence Klein tunnelling and bilayer anti-Klein reflection

- **Orbital Hall Effect** Presentation *(PH 557)* *(Aug '25 – Nov '25)*
  Prof. Sayantika Bhowal | Department of Physics, IIT Bombay
  - Reviewed the theory and experiment of the orbital Hall effect in a five-member team, from two-orbital tight-binding models and orbital Berry curvature to its 2023 observation in titanium
  - Presented the quantum and spin Hall effects, their chiral and helical edge states, and the time-reversal-symmetry comparison that separates them from the non-topological orbital Hall effect

- **Orbital Dynamics of Similar and Dissimilar Objects** *(PH 222)* *(Jan '25 – Apr '25)*
  Prof. Pradeep Sarin | Department of Physics, IIT Bombay
  - Built an Arduino Mega–TFT display system to simulate binary-star and satellite–planet orbital trajectories
  - Validated simulated trajectories, energies, and periods against analytical results to within 0.02% error

- **Nonlinear Dynamics of the Chaotic Duffing Oscillator** Report *(PH 567)* *(Aug '24 – Dec '24)*
  Prof. Punit Parmananda | Department of Physics, IIT Bombay

- Simulated driven Duffing dynamics across periodic, bifurcating, and chaotic regimes under varied forcing
- Analyzed fixed-point stability via time series, phase portraits, Poincaré sections and an analog circuit

- **Physics-Informed Quantum System Analysis** Report *(PH 227)* *(Aug '24 – Dec '24)*
  Prof. Alok Shukla & Prof. Sadhana Dash | Department of Physics, IIT Bombay
  - Generated Schrödinger datasets and trained PINNs to predict quantum harmonic-oscillator wavefunctions
  - Achieved an MSE of 0.00077, beating random forests (0.01295) and XGBoost (0.03213) on identical data
- **Voice and Song Recognition** *(DS 203)* *(Aug '24 – Dec '24)*
  Prof. Vinay Kulkarni | CMInDS, IIT Bombay
  - Extracted MFCC features from 100+ vocal clips to identify singers, songs, and musical styles
  - Benchmarked SVM, k-NN, and random-forest classifiers for multiclass recognition across the dataset

Technical Skills

- **Experimental:** Bluefors dilution refrigerator, SET charge sensing, RF reflectometry, Zurich Instruments/Qblox hardware, VNA, lock-in amplifier, AWG, source-measure units, oscilloscope
- **Programming:** Python, C++, MATLAB, Java, JavaScript
- **Scientific Computing & ML:** QCoDeS, Z2Pack, NumPy, SciPy, Matplotlib, Pandas, PyTorch, scikit-learn, XGBoost
- **Software & Tools:** LaTeX, LTspice, PHOEBE, ReactJS, HTML/CSS, Figma, Blender, Photoshop

Teaching and Mentorship

- **Teaching Assistant, Introduction to Quantum Mechanics**: tutorials for 30 undergraduates *(Jan '26 – Apr' 26)*
- **Teaching Assistant, Electronics Laboratory**: lab sessions for 60+ Master's students *(Aug '25 – Nov '25)*
- **Mentor, Seasons of Code**: six-week quantum computing program for 13 students *(May '25 – Jul '25)*

Key Courses Undertaken

- **Physics**: Quantum Mechanics I & II, Statistical Mechanics, Electromagnetic Theory, Condensed Matter Physics, Magnetism and Superconductivity, Quantum Information and Computing, Computational Many-Body Physics
- **Mathematics**: Calculus, Linear Algebra, Differential Equations, Complex Analysis, Integral Transforms
- **Other**: AI and Data Science, Computer Programming, Analog and Digital Electronics, Economics, Psychology

Extracurricular Activities

Vedic Math	• Won top honors twice in district-level Vedic mathematics competitions
Social	• Contributed 80+ hours to NSS Green Campus through campus tree-plantation drives
Sports	• Participated in the Inter-IIT Cricket Tournament, where the team won a bronze medal
Hobbies	• Participate in quizzes with the IIT Bombay Quizzing Club

References

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